222651: metadacite, Mount Menzies

(Cock Robin Formation, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Menzies (SH 51-5), Menzies (3138)
MGA Zone 51, 313626E 6715605N

Sampled on 17 April 2018

This sample was collected from a trench on Jeedamya Station, about 3.4 km west-southwest of Oliver Twist Well, 2.0 north-northwest of Mount Menzies, and 1.7 km southeast of the Goldfields Highway.

Geological context

The unit sampled is a metadacite of the Cock Robin Formation of the west-central Kalgoorlie Terrane (Sapkota, 2017). The formation consists mainly of ultramafic schist, fuchsite–quartz–andalusite schist, komatiitic basalt, basalt, amphibolite, slate, and banded chert. The formation is similar lithologically to the Trevors Bore and Mount Leonora Formations to the north (Sapkota, 2017), and was deformed together with the Twin Hills Monzogranite, samples of which 27 km to the north yielded igneous crystallization ages of 2803 ± 3 Ma (GA 2001967041A, Dunphy et al., 2003) and 2801 ± 3 Ma (GSWA 222655, Wingate et al., 2021). The present sample was collected to determine the age of the Cock Robin Formation.

Petrographic description

The sample is a schistose metadacite, consisting of about 75% quartzofeldspathic groundmass, 15% plagioclase phenocrysts, 5% secondary quartz, 5% secondary muscovite and trace tourmaline and rutile. The groundmass is a finely recrystallized aggregate of quartz and feldspar with a well-developed polygonal texture and an average grain size of 0.05 mm. Plagioclase phenocrysts are subhedral to euhedral, mainly tabular, up to 2.5 mm long, lightly to moderately sericitized, and albitized, with compositions between An_6 and An_{10}. The groundmass also contains lenses of more coarsely recrystallized quartz and feldspar, mainly as pressure shadows around some plagioclase phenocrysts. Muscovite laths are up to 0.15 mm long, well aligned, and form either single grains or distinct folia. Tourmaline is anhedral, elongate, micropoikilitic, almost colourless to olive green, and up to 0.7 mm long. Dark brown anhedral grains of probable rutile are up to 0.2 mm in size. The sample contains a tightly folded quartz vein about 1 mm wide, and the axial planes of the folds parallel the foliation and contain more coarsely crystalline muscovite laths. The sample has been deformed and recrystallized at relatively low metamorphic grade, probably greenschist facies.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and subhedral to euhedral. The crystals are up to 250 µm long, and mainly elongate, with aspect ratios up to 6:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and most crystals are dominated by high-uranium, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 June 2020, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.70% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.27% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.39%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). One analysis is >30% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and considered not to be geologically significant. The remaining 17 analyses form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 17 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2804 ± 6 Ma (MSWD = 0.70). A discordia regression through data in Group I yields an upper intercept of 2805 ± 8 Ma and a lower intercept within uncertainty of 0 Ma (MSWD = 0.74), indicating that loss of radiogenic Pb was geologically recent.

The date of 2804 ± 6 Ma for the 17 analyses in Group I is interpreted as the igneous crystallization age of the dacite.

Figure 1. Hand specimen image for sample 222651: metadacite, Mount Menzies

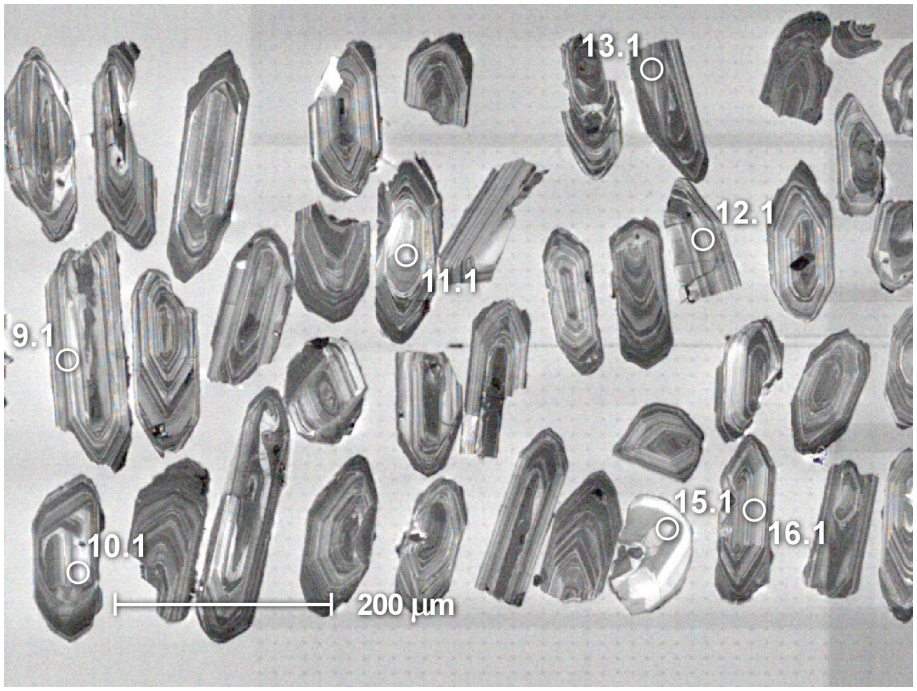

Figure 2. Cathodoluminescence image of representative zircons from sample 222651: metadacite, Mount Menzies. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 222651: metadacite, Mount Menzies

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	7	7.1	52	12	0.23	0.243	1.916	0.036	0.19612	0.00188	1.921	0.036	0.19396	0.00208	2702	42	2776	18	2.7
I	15	15.1	61	27	0.44	0.140	1.879	0.034	0.19737	0.00178	1.882	0.034	0.19613	0.00188	2747	41	2794	16	1.7
I	5	5.1	155	78	0.50	0.000	1.872	0.024	0.19664	0.00119	1.872	0.024	0.19664	0.00119	2759	29	2799	10	1.4
I	2	2.1	115	44	0.38	-0.017	1.833	0.026	0.19734	0.00131	1.833	0.026	0.19750	0.00131	2806	32	2806	11	0.0
I	1	1.1	79	26	0.34	0.077	1.846	0.030	0.19898	0.00157	1.848	0.030	0.19829	0.00162	2789	37	2812	13	0.8
I	17	17.1	133	61	0.46	0.000	1.868	0.037	0.19838	0.00126	1.868	0.037	0.19838	0.00126	2764	45	2813	10	1.7
I	6	6.1	44	14	0.32	-0.146	1.831	0.038	0.19815	0.00207	1.828	0.038	0.19944	0.00220	2813	47	2822	18	0.3
I	10	10.1	116	29	0.25	0.142	2.570	0.036	0.19652	0.00148	2.573	0.036	0.19525	0.00157	2116	25	2787	13	24.1
I	13	13.1	199	91	0.46	0.081	2.400	0.029	0.19685	0.00184	2.402	0.029	0.19614	0.00187	2244	23	2794	16	19.7
I	12	12.1	113	39	0.35	-0.039	2.006	0.029	0.19587	0.00136	2.005	0.029	0.19622	0.00139	2608	31	2795	12	6.7
I	14	14.1	209	81	0.39	0.413	2.794	0.034	0.19993	0.00125	2.805	0.034	0.19626	0.00144	1965	21	2795	12	29.7
I	4	4.1	174	89	0.51	0.274	2.423	0.030	0.19930	0.00128	2.429	0.031	0.19686	0.00141	2222	24	2800	12	20.6
I	11	11.1	66	31	0.48	0.034	2.033	0.036	0.19747	0.00175	2.033	0.036	0.19717	0.00177	2578	37	2803	15	8.0
I	9	9.1	157	71	0.45	0.414	2.878	0.037	0.20121	0.00149	2.890	0.038	0.19754	0.00173	1915	22	2806	14	31.7
I	18	18.1	159	56	0.35	0.065	2.202	0.051	0.19826	0.00128	2.203	0.051	0.19768	0.00131	2412	46	2807	11	14.1
I	16	16.1	212	74	0.35	0.000	2.671	0.032	0.19847	0.00120	2.671	0.032	0.19847	0.00120	2050	21	2814	10	27.1
I	8	8.1	158	81	0.51	-0.086	1.981	0.026	0.19812	0.00121	1.979	0.026	0.19888	0.00125	2637	28	2817	10	6.4
D	3	3.1	349	174	0.50	0.011	3.374	0.036	0.20840	0.00113	3.374	0.036	0.20830	0.00114	1673	16	2892	9	42.1

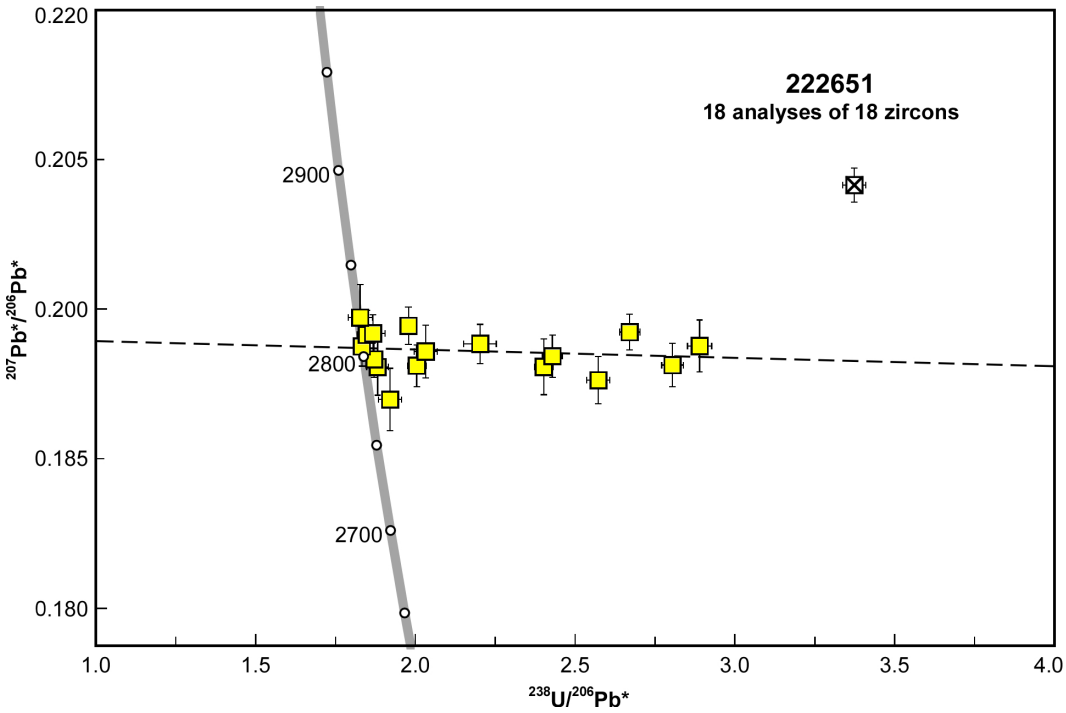

Figure 3. **U–Pb analytical data for zircons from sample 222651: metadacite, Mount Menzies. Yellow squares indicate Group I (magmatic zircons); crossed square indicates Group D (discordance >30%). The dashed line indicates a discordia regression through data in Group I**

References

Dunphy, JM, Fletcher, IR, Cassidy, KF and Champion, DC 2003, Compilation of SHRIMP U-Pb geochronological data, Yilgarn Craton, Western Australia, 2001-2002: Geoscience Australia, Record 2003/15, 139p.

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Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and Sapkota, J 2022, 222651: metadacite, Mount Menzies; Geochronology Record 1853: Geological Survey of Western Australia, 4p.

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